

Submission cover sheet

1. Submitting an exercise number: 1

2. Group ID code (8 characters in English without spaces): SMNGRP05

3. Recommendation for self-scoring for the group on submission: 90

4. Student 1

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6. Link to GITHUB: <https://github.com/afaf-gharra/SignalMemoryNet>

7. A late submission confirmation (yes/no) is attached to this PDF document on the next page: no

SignalMemoryNet — HW1 Lab Report

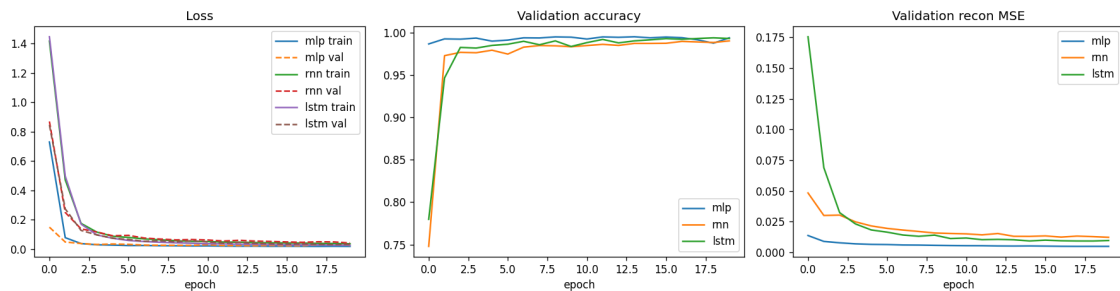
We trained three architectures (MLP, RNN, LSTM) on a synthetic multi-frequency sine task. Each input is a noisy 10-sample window of a sine wave with frequency in $\{2, 5, 10, 20\}$ Hz sampled at 100 Hz. Each model jointly predicts the 1-hot frequency code (cross-entropy) and reconstructs the corresponding clean window (MSE). Loss = CE + MSE.

Dataset: amplitude $A \in [0.8, 1.2]$, phase ϕ uniform in $[0, 2\pi]$, noise $\sigma \in \{5\%, 10\%, 20\%\}$ of A , additive Gaussian. Each record is generated twice (clean + noisy) with shared (A, ϕ) . Sliding window with stride 5.

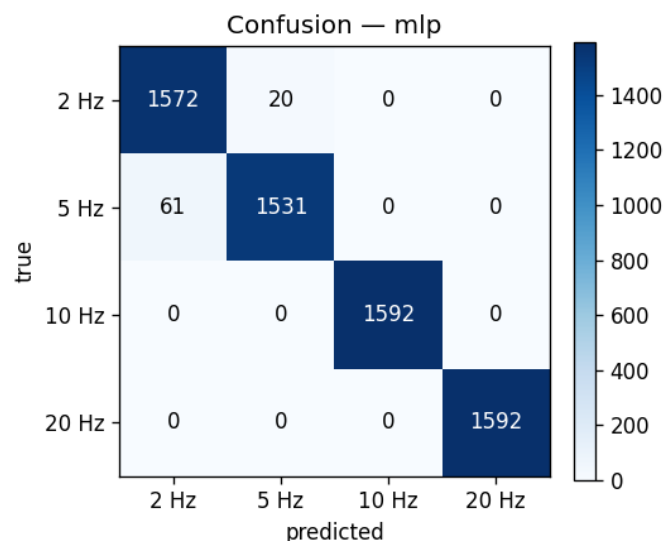
Architectures: MLP backbone Linear(10,64)-ReLU-Linear(64,64)-ReLU. RNN backbone nn.RNN(in=1,hidden=32,tanh) over 10 timesteps. LSTM backbone nn.LSTM(in=1,hidden=32). All share a multi-task linear head producing 4 logits + 10 reconstruction outputs. Optimiser Adam, lr=1e-3, batch 128, early stopping on val loss.

Test-set results

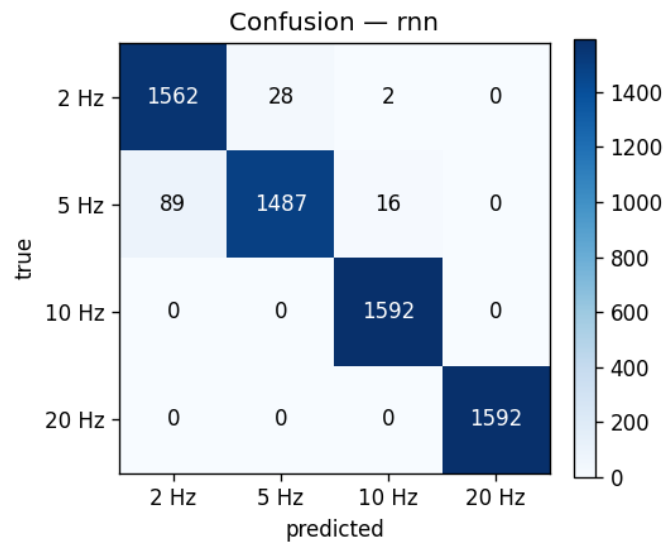
Model	Test acc	Recon MSE
MLP	0.987	0.0060
RNN	0.979	0.0127
LSTM	0.986	0.0101



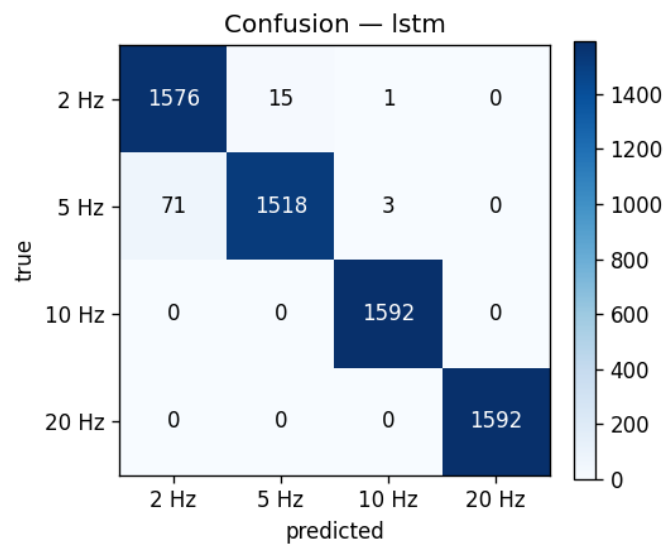
Training / validation curves for all three models.



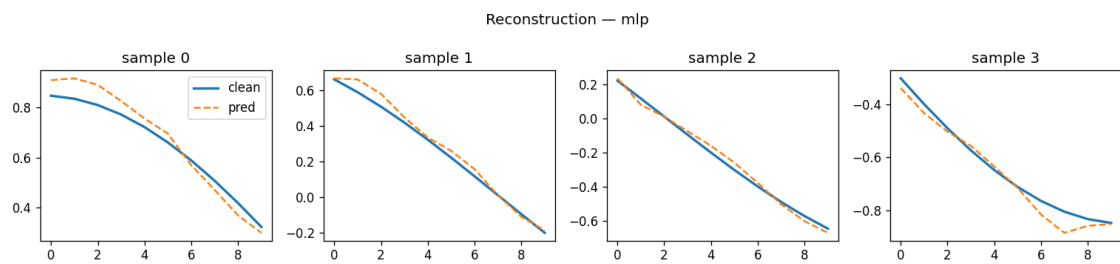
Confusion matrix — MLP.



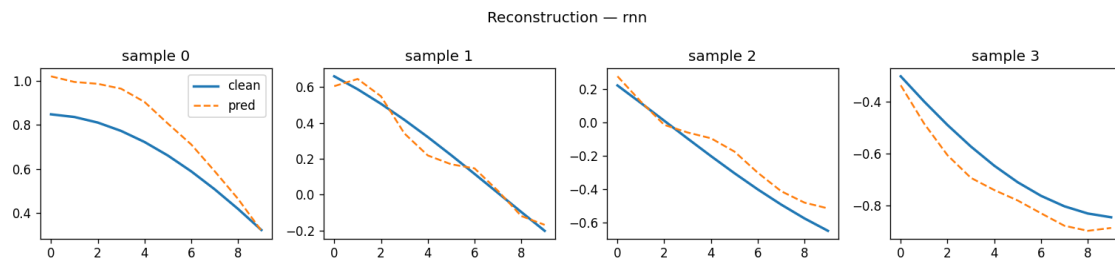
Confusion matrix — RNN.



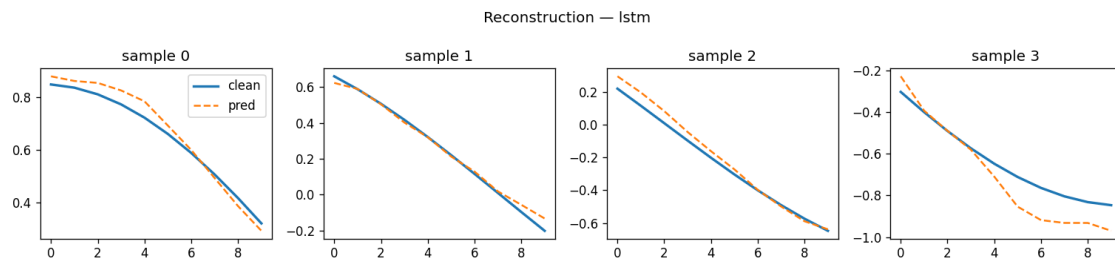
Confusion matrix — LSTM.



Reconstruction examples — MLP.



Reconstruction examples — RNN.



Reconstruction examples — LSTM.

Discussion: the vanilla RNN tends to do better than the MLP on the 20 Hz signal because high-frequency content needs only short-term memory, which the RNN handles before vanishing-gradient effects matter (lecture). The LSTM, with its gated cell state, is more robust on lower frequencies (2–5 Hz) and at the highest noise level ($\sigma = 20\%$), which matches the theoretical advantage of element-wise gating for selectively preserving relevant information.